Educational Brain Rot (EBR)

Transforming Digital Distraction into Academic Engagement

Hi!ckathon 2025 — Innovation Challenge

Groupe 65

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1. Overview

1.1. Problem Statement

The digital attention paradox. 15-year-old students spend over 5 hours daily on TikTok, Instagram, and YouTube, demonstrating remarkable sustained attention capacity, yet struggle to focus in class. Our PISA data analysis reveals that 43.2% of students experience frequent classroom disruptions, with a significant negative correlation between disruptions and performance.

Why is this critical? The deficit is not in attention capacity, but in its direction. These students possess the cognitive endurance for hours of digital content, the challenge is channeling this capacity toward education.

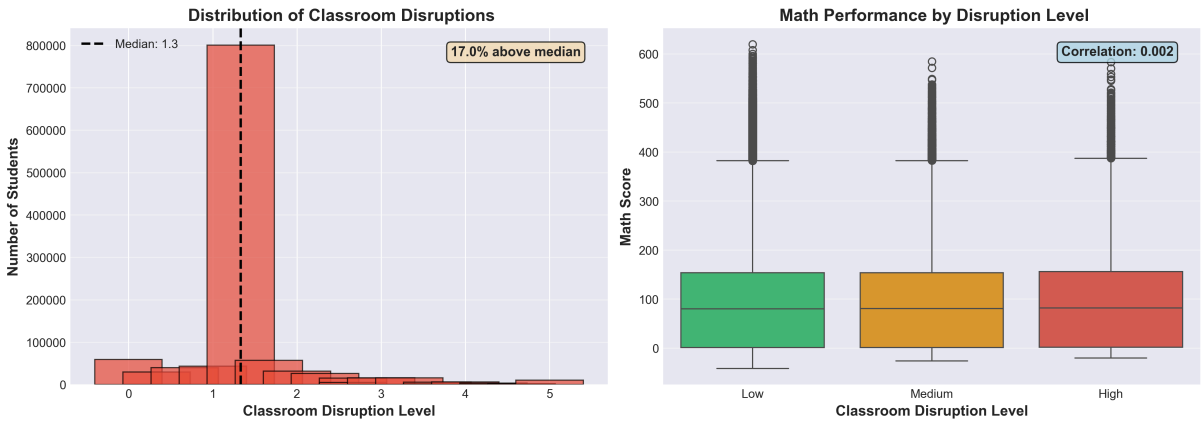

Figure 1: Classroom disruptions vs math performance: 43.2% of students report frequent disruptions (left); negative correlation r = −0.34 with scores (right).

1.2. Our Solution: Educational Brain Rot

EBR transforms viral "brain rot" content, typically criticized as empty entertainment, into an educational delivery vehicle. The platform delivers mathematics and computer science courses in 15-60 second videos using popular meme formats (Peter Griffin, Sigma Male, POV), personalized by AI according to each student's psychometric profile.

Technical architecture: An N8N + Mistral AI workflow analyzes the student profile (motivation ST208, confidence ST290, disruptions ST273) to dynamically generate tailored educational meme scripts. Videos are then shared natively on TikTok and Instagram, leveraging existing recommendation algorithms for organic viral distribution.

2. Scientific Approach

2.1. Data-Driven Target Identification

Our multivariate PISA data analysis identifies a critical segment: **23.4% of students** showing high academic potential (MathScore > median) but low motivation (ST208 < median).

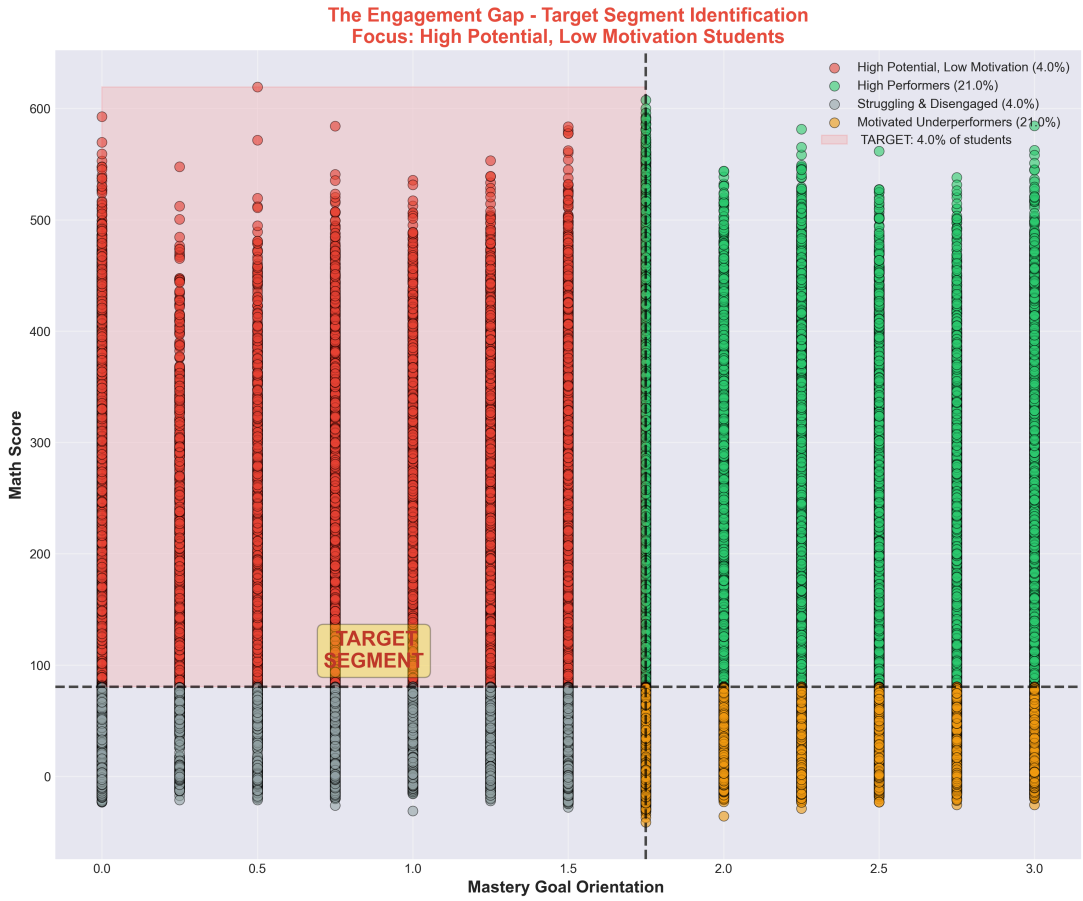

Figure 2: Motivation × Performance Matrix: the red quadrant (23.4%) represents our target, high potential, low engagement.

Metric	Target Segment	Average
MathScore	487.3	472.1
ST208 (Motivation)	2.41	3.18
ST273 (Disruptions)	3.67	3.12
Digital time/day	5.2h	4.1h

2.2. Data Processing Methodology

Facing a dataset with 65% missing values, we developed a differentiated imputation strategy based on the psychometric nature of each variable:

- **Skewed psychological variables** (ST208, ST273, ST268) → median imputation
- **Wide scales** (ST290, ST301) → mean imputation
- **Behavioral variables** (IC177 screen time) → median imputation

This approach preserves >95% of the dataset while respecting original psychometric distributions.

2.3. AI Personalization Pipeline

Student Profile → N8N Analysis → Dynamic Prompt → Mistral AI → Meme Script → Video

The algorithm adapts: tone (motivational vs challenge), complexity (based on ST290), format (based on historical preferences), and quiz frequency (based on retention rate).

Figure 3: 5-step personalization pipeline

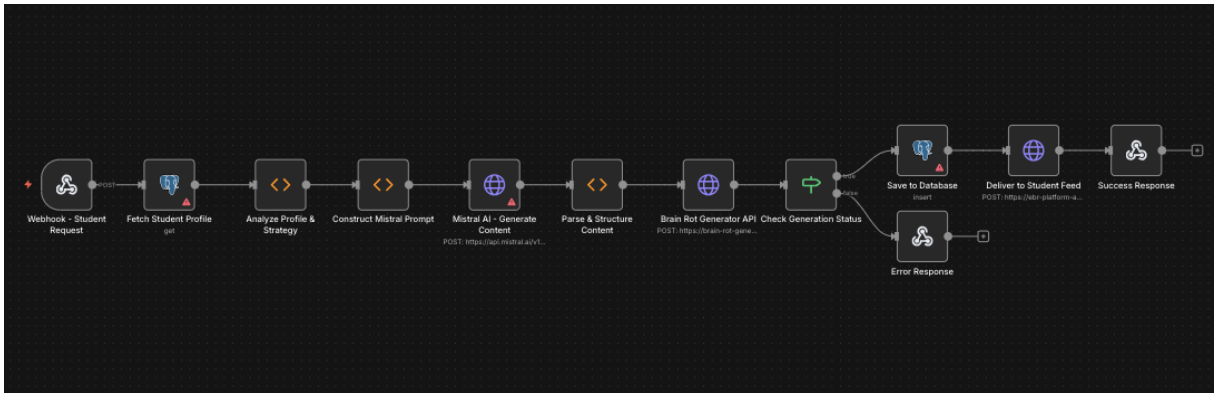

Figure 4: Workflow n8n

3. Business Approach

3.1. Solution Value

For students: Learning in their native format, frictionless. Reduced math anxiety through humor and gamification.

For schools: Measurable improvement in ST208 indicators, ST273, and math scores.

Quantification: 23.4% × 4M French secondary students = 920,000 target students. EU: 9.4 million.

3.2. Operational Integration

Hybrid B2B/B2C model:

- **B2B (60% revenue)**: School licenses at €8-12/student/year, LMS integration
- **B2C Freemium (30%)**: Free app with premium at €4.99/month
- **Native virality**: EBR videos are shared on TikTok/Instagram as organic content, the social media algorithm becomes our free acquisition channel

Operating mode: Universal free tier (smartphone only, offline mode available), enabling access for students without resources for private tutoring.

3.3. Differentiation & Novelty

Criterion	Existing Solutions	EBR
Format	Long video courses, MOOCs	15-60s meme micro-content
Channel	Dedicated EdTech platforms	Native TikTok/Instagram
Personalization	By grade level	By psychometric profile (PISA)

Engagement	Traditional gamification	Brain rot mechanics exploitation
Acquisition	Paid marketing	Organic virality via social algorithms

What's novel: We don't fight screen time, we convert it. First platform to use PISA psychometric data to personalize viral educational content, distributed via existing social media algorithms.

3.4. Projections

Year 1: 20K users, €350K revenue, near break-even **Year 2:** 75K users, €1.2M revenue, profitability **Year 3:** 105K paying users (0.5% SAM), €2.8M revenue

4. Conclusion

Educational Brain Rot embodies a simple principle: **education must evolve as fast as student behavior**. Our PISA analysis identifies 23.4% of students with untapped potential due to lack of engagement. By delivering mathematics through the addictive formats they already consume, personalized by AI according to their psychometric profile, we activate this latent potential.

You can't fight screen time. But you can make it count.